Chapter 13. Classification of Covering Spaces

Equivalence of coverings, universal covering space, covering transformations, existence of coverings

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Throughout this chapter a covering map $p : E \to B$ is assumed to have E and B path connected and locally path connected. For $p(e_0) = b_0$ let $H_0 = p_*(\pi_1(E, e_0))$, a subgroup of $\pi_1(B, b_0)$ isomorphic to $\pi_1(E, e_0)$ (Theorem 9.17).

13.1 Equivalence of covering spaces

Motivation. Covering spaces were introduced to compute fundamental groups; this chapter goes the other way and classifies covering spaces by subgroups of π_1, in close analogy with Galois theory, where intermediate fields correspond to subgroups.

Covering maps $p : E \to B$ and $p' : E' \to B$ are **equivalent** if there is a homeomorphism $h : E \to E'$ with $p = p' \circ h$ (an **equivalence**).

Lemma 13.1 (79.1, general lifting lemma)

Let $p : E \to B$ be a covering map, $p(e_0) = b_0$, and $f : Y \to B$ continuous with $f(y_0) = b_0$, where Y is path connected and locally path connected. Then f has a lifting $\tilde{f} : Y \to E$ with $\tilde{f}(y_0) = e_0$ iff $f_*(\pi_1(Y, y_0)) \subset p_*(\pi_1(E, e_0))$. Such a lifting is unique.

Proof. If $\tilde{f}$ exists, $f_* = p_*\tilde{f}_*$.

Uniqueness: for $y_1 \in Y$ and a path α from y_0 to y_1, $\tilde{f} \circ \alpha$ is the lifting of $f \circ \alpha$ from e_0, so $\tilde{f}(y_1)$ is its end point.

Existence: define $\tilde{f}(y_1)$ as the end point of the lifting of $f \circ \alpha$ from e_0. It is well defined: if β is another path from y_0 to y_1, lift $f \circ \alpha$ to γ from e_0 and $f \circ \bar{\beta}$ to δ from $\gamma(1)$; then $\gamma * \delta$ lifts the loop $f \circ (\alpha * \bar{\beta})$, whose class lies in $f_*\pi_1(Y, y_0) \subset H_0$, so it is a loop (Theorem 9.17(c)); then $\bar{\delta}$ is the lifting of $f \circ \beta$ from e_0 and ends at $\gamma(1)$. Continuity at y_1: given a neighbourhood N of $\tilde{f}(y_1)$, choose a path-connected evenly covered $U \ni f(y_1)$ whose slice V_0 through $\tilde{f}(y_1)$ lies in N, and a path-connected neighbourhood W of y_1 with $f(W) \subset U$. For $y \in W$ and a path δ in W from y_1 to y, the lifting of $f \circ (\alpha * \delta)$ is $\gamma * ((p|V_0)^{-1} \circ f \circ \delta)$, ending in V_0. So $\tilde{f}(W) \subset V_0 \subset N$. ∎

Theorem 13.2 (79.2)

Let $p : E \to B$, $p' : E' \to B$ be covering maps with $p(e_0) = p'(e_0') = b_0$. There is an equivalence $h : E \to E'$ with $h(e_0) = e_0'$ iff $H_0 = p_*\pi_1(E, e_0)$ and $H_0' = p_*'\pi_1(E', e_0')$ are equal. Such h is unique.

Proof. If h exists, h_* is an isomorphism and $p = p'h$ gives $H_0 = H_0'$. Conversely, Lemma 13.1 gives liftings $h : E \to E'$ of p with $h(e_0) = e_0'$ and $k : E' \to E$ of p' with $k(e_0') = e_0$. Then $k \circ h$ and id_E both lift p and fix e_0, so they are equal by uniqueness; likewise $h \circ k = \mathrm{id}$. Uniqueness of h is that of liftings. ∎

Lemma 13.3 (79.3)

Let $e_0, e_1 \in p^{-1}(b_0)$ and $H_i = p_*\pi_1(E, e_i)$.
(a) If γ is a path in E from e_0 to e_1 and $\alpha = p \circ \gamma$, then $[\alpha]H_1[\alpha]^{-1} = H_0$.
(b) Conversely, if H is conjugate to H_0, then $H = H_1$ for some $e_1 \in p^{-1}(b_0)$.

Proof. (a) If $[h] \in H_1$, $h = p \circ \tilde{h}$ with $\tilde{h}$ a loop at e_1; $\gamma * \tilde{h} * \bar{\gamma}$ is a loop at e_0 mapping to $[\alpha][h][\alpha]^{-1}$, so $[\alpha]H_1[\alpha]^{-1} \subset H_0$. Using $\bar{\gamma}$, $[\alpha]^{-1}H_0[\alpha] \subset H_1$.

(b) If $H_0 = [\alpha]H[\alpha]^{-1}$, lift α to γ from e_0 and let $e_1 = \gamma(1)$; by (a), $H_0 = [\alpha]H_1[\alpha]^{-1}$, so $H = H_1$. ∎

Theorem 13.4 (79.4)

p and p' are equivalent iff H_0 and H_0' are conjugate in $\pi_1(B, b_0)$.

Proof. If h is an equivalence and $e_1' = h(e_0)$, then $H_0 = p_*'\pi_1(E', e_1')$ by Theorem 13.2, which is conjugate to H_0' by Lemma 13.3. Conversely, if they are conjugate, Lemma 13.3(b) gives e_1' with $p_*'\pi_1(E', e_1') = H_0$, and Theorem 13.2 gives an equivalence carrying e_0 to e_1'. ∎

Example 13.1

$\pi_1(S^1) \cong \mathbb{Z}$ is abelian with subgroups $n\mathbb{Z}$. So every covering of S^1 is equivalent to $p : \mathbb{R} \to S^1$ (trivial subgroup) or to $z \mapsto z^n$ (subgroup $n\mathbb{Z}$).

13.2 The universal covering space

Motivation. The simply connected covering space sits above all others, just as the splitting field sits above every intermediate field. Its existence, under mild local conditions, makes the correspondence complete.

If E is simply connected, p is a **universal covering**; by Theorem 13.4 it is unique up to equivalence.

Lemma 13.5 (80.1)

Let B be path connected and locally path connected, and $p : E \to B$ a covering map with E not necessarily path connected. If E_0 is a path component of E, then $p|E_0 : E_0 \to B$ is a covering map.

Proof. E is locally path connected, so E_0 is open and $p(E_0)$ is open. If $x \in \overline{p(E_0)}$, take a path-connected evenly covered $U \ni x$; some slice V meets E_0, and being path connected lies in E_0, so $x \in U = p(V) \subset p(E_0)$. Thus $p(E_0) = B$. For any such U, $(p|E_0)^{-1}(U)$ is the union of the slices meeting E_0, each contained in E_0 and mapped homeomorphically onto U. ∎

Lemma 13.6 (80.2)

Let $p = r \circ q$ with $p : X \to Z$, $q : X \to Y$, $r : Y \to Z$ continuous (all spaces path connected and locally path connected).
(a) If p and r are covering maps, so is q.
(b) If p and q are covering maps, so is r.

Proof. Let $x_0 \in X$, $y_0 = q(x_0)$.

(a) q is surjective: for a path $\tilde{\alpha}$ in Y from y_0 to y, lift $r \circ \tilde{\alpha}$ through p to $\tilde{\tilde{\alpha}}$ from x_0; $q \circ \tilde{\tilde{\alpha}}$ lifts $r \circ \tilde{\alpha}$ through r from y_0, so equals $\tilde{\alpha}$, and y is in the image. Given y, take a path-connected $U \ni r(y)$ evenly covered by both p and r, and the slice V of $r^{-1}(U)$ containing y. Each slice U_α of $p^{-1}(U)$ is connected, so q maps it into one slice of $r^{-1}(U)$; $q^{-1}(V)$ is the union of those U_α mapped into V, and on each $q = (r|V)^{-1} \circ (p|U_\alpha)$ is a homeomorphism onto V.

(b) Let U be a path-connected neighbourhood of z evenly covered by p, and V_β the path components of $r^{-1}(U)$, open and disjoint. The slices U_α of $p^{-1}(U)$ are the path components of $p^{-1}(U)$, and q maps each into one V_β; so $q^{-1}(V_\beta)$ is a union of slices and, q being surjective, contains some U_α. By Theorem 9.10, q restricts to a covering map $q^{-1}(V_\beta) \to V_\beta$, and U_α is a path component of $q^{-1}(V_\beta)$, so $q|U_\alpha : U_\alpha \to V_\beta$ is surjective by Lemma 13.5. It is also open and injective (as $p|U_\alpha$ is injective).

Thus $q|U_\alpha$ is a homeomorphism onto V_β, and $r|V_\beta = (p|U_\alpha)\circ(q|U_\alpha)^{-1}$ is a homeomorphism onto U. ■

Theorem 13.7 (80.3)

If $p : E \to B$ is a covering map with E simply connected, then for any covering map $r : Y \to B$ there is a covering map $q : E \to Y$ with $r \circ q = p$.

Proof. Lemma 13.1 applies trivially and gives q with $r\circ q = p$; Lemma 13.6(a) shows q is a covering map. ■

Lemma 13.8 (80.4)

If $p : E \to B$ is a covering map with E simply connected and $p(e_0) = b_0$, then b_0 has a neighbourhood U such that $i_* : \pi_1(U, b_0) \to \pi_1(B, b_0)$ is trivial.

Proof. Let U be evenly covered and $\tilde{U}$ the slice containing e_0. A loop f in U lifts to a loop in $\tilde{U}$, which is nulhomotopic in E; applying p gives a path homotopy in B from f to a constant. ■

Example 13.2

The infinite earring X (Example 11.2) is path connected and locally path connected but has no universal covering: for any neighbourhood U of the origin, some circle C_n lies in U, and C_n is a retract of X, so $\pi_1(C_n) \to \pi_1(X)$ is injective and $i_* : \pi_1(U) \to \pi_1(X)$ is nontrivial.

13.3 Covering transformations

Motivation. The symmetries of a covering (deck transformations) form a group that, for the universal cover, is isomorphic to $\pi_1(B)$. This is the topological counterpart of the Galois group.

An equivalence of p with itself is a **covering transformation**; they form the group $\mathcal{C}(E, p, B)$. The **normalizer** of H in G is $N(H) = \{g : gHg^{-1} = H\}$, the largest subgroup in which H is normal. Let $F = p^{-1}(b_0)$, $\Phi : \pi_1(B, b_0)/H_0 \to F$ the bijection induced by the lifting correspondence (Theorem 9.17), and $\Psi : \mathcal{C}(E, p, B) \to F$, $\Psi(h) = h(e_0)$, injective by uniqueness of liftings.

Lemma 13.9 (81.1)

The image of Ψ equals $\Phi(N(H_0)/H_0)$.

Proof. Let γ be the lifting of a loop α from e_0, $e_1 = \gamma(1) = \Phi([\alpha]H_0)$, $H_1 = p_*\pi_1(E, e_1)$. A covering transformation with $h(e_0) = e_1$ exists iff $H_1 = H_0$ (Theorem 13.2). Since $[\alpha]H_1[\alpha]^{-1} = H_0$ (Lemma 13.3), this holds iff $[\alpha]H_0[\alpha]^{-1} = H_0$, i.e. $[\alpha] \in N(H_0)$. ■

Theorem 13.10 (81.2)

$\Phi^{-1}\circ\Psi : \mathcal{C}(E, p, B) \to N(H_0)/H_0$ is a group isomorphism.

Proof. It is a bijection by Lemma 13.9. Let $h(e_0) = e_1$, $k(e_0) = e_2$, and choose paths γ, δ from e_0 to e_1, e_2; with $\alpha = p\circ\gamma$, $\beta = p\circ\delta$, $\Phi([\alpha]H_0) = e_1$ and $\Phi([\beta]H_0) = e_2$. Then $h\circ\delta$ runs from e_1 to $e_3 = h(k(e_0))$ and lifts β, so $\gamma * (h\circ\delta)$ lifts $\alpha*\beta$ from e_0 and ends at e_3: $\Phi([\alpha*\beta]H_0) = \Psi(h\circ k)$. ■

Corollary 13.11 (81.3)

H_0 is normal in $\pi_1(B, b_0)$ iff for all $e_1, e_2 \in p^{-1}(b_0)$ some covering transformation carries e_1 to e_2. Then $\mathcal{C}(E, p, B) \cong \pi_1(B, b_0)/H_0$.

Proof. If H_0 is normal, Ψ is surjective, giving h_i with $h_i(e_0) = e_i$; take $h_2 h_1^{-1}$. Conversely, transitivity makes Ψ surjective, so $N(H_0)/H_0 = \pi_1/H_0$ and $N(H_0) = \pi_1(B, b_0)$. ■

Corollary 13.12 (81.4)

If E is simply connected, $\mathcal{C}(E, p, B) \cong \pi_1(B, b_0)$.

If H_0 is normal, p is **regular**. Then $\mathcal{C}$ is transitive on every fibre $p^{-1}(b)$: for $x_1, x_2 \in p^{-1}(b)$ and a path β from b to b_0, lift β from x_1, x_2 to paths ending at e_1, e_2, and let $h(e_1) = e_2$; then h carries the first lifting to a lifting of β ending at e_2, which by uniqueness (lift $\bar{\beta}$ from e_2) is the second, so $h(x_1) = x_2$.

Example 13.3

Every covering of S^1 is regular; for $\mathbb{R} \to S^1$ the covering transformations are $x \mapsto x + n$. For the covering of the figure eight in Lemma 9.45, $\mathcal{C}$ is trivial: a loop around A lifts to a non-loop from the origin but to a loop from every other point of the fibre on the y-axis, and similarly for B, so every covering transformation fixes the origin.

For a group G of homeomorphisms of X, the **orbit space** X/G is the quotient by $x \sim g(x)$. The action is **properly discontinuous** if each x has a neighbourhood U with $g(U) \cap U = \emptyset$ for $g \neq e$; then $g_0(U) \cap g_1(U) = \emptyset$ for $g_0 \neq g_1$.

Theorem 13.13 (81.5)

Let X be path connected and locally path connected and G a group of homeomorphisms of X. The quotient map $\pi : X \to X/G$ is a covering map iff the action is properly discontinuous. In that case π is regular and G is its group of covering transformations.

Proof. π is open, since $\pi^{-1}\pi(U) = \bigcup_g g(U)$. If the action is properly discontinuous and U is as in the definition, $\pi^{-1}\pi(U)$ is the disjoint union of the open sets $g(U)$, each meeting each orbit at most once; $\pi | g(U)$ is a continuous open bijection onto $\pi(U)$, hence a homeomorphism. Conversely, if π is a covering, let U be the slice through x over an evenly covered neighbourhood of $\pi(x)$; if $g \neq e$ and $u' = g(u) \in U$ with $u \in U$, then $u' = u$ by injectivity of $\pi | U$; so g fixes u, and g is a covering transformation fixing a point, hence the identity (shown next), a contradiction. Each g is a covering transformation. If h is one with $h(x_1) = x_2$, then x_1, x_2 lie in one orbit, $g(x_1) = x_2$ for some g, and $h = g$ by Theorem 13.2. G is transitive on fibres, so π is regular (Corollary 13.11). ■

Theorem 13.14 (81.6)

If $p : X \to B$ is regular with group of covering transformations G, there is a homeomorphism $k : X/G \to B$ with $p = k \circ \pi$.

Proof. p is constant on orbits, so it induces a continuous k. Conversely p is a quotient map (continuous, open, surjective), and π is constant on each fibre of p since G is transitive on fibres; so π induces a continuous inverse $B \to X/G$. ■

Example 13.4

On the cylinder $X = S^1 \times I$, $h(x,t) = (-x,t)$ and $k(x,t) = (-x, 1-t)$ generate properly discontinuous actions of $\mathbb{Z}/2$; $X/\langle h\rangle$ is again a cylinder, $X/\langle k\rangle$ is the Möbius band. $\mathbb{Z}^2$ acting on $\mathbb{R}^2$ by translations gives the torus; $\{\pm 1\}$ acting on S^n gives P^n.

13.4 Existence of covering spaces

Motivation. Every subgroup of $\pi_1(B)$ is realized by a covering space, provided B is semilocally simply connected. This completes the classification.

B is **semilocally simply connected** if each b has a neighbourhood U such that $\pi_1(U,b) \to \pi_1(B,b)$ is trivial. By Lemma 13.8 this is necessary for a universal covering.

Theorem 13.15 (82.1)

Let B be path connected, locally path connected and semilocally simply connected, $b_0 \in B$, and H a subgroup of $\pi_1(B,b_0)$. There is a covering map $p : E \to B$ and $e_0 \in p^{-1}(b_0)$ with $p_*\pi_1(E,e_0) = H$.

Proof.

Step 1. Let $\mathcal{P}$ be the set of paths in B from b_0. Put $\alpha \sim \beta$ if $\alpha(1) = \beta(1)$ and $[\alpha * \bar\beta] \in H$, an equivalence relation; write $\alpha^\#$ for the class, E for the set of classes, and $p(\alpha^\#) = \alpha(1)$. Note:

(1) $[\alpha] = [\beta]$ implies $\alpha^\# = \beta^\#$;

(2) $\alpha^\# = \beta^\#$ implies $(\alpha * \delta)^\# = (\beta * \delta)^\#$ for any path δ from $\alpha(1)$, since $[(\alpha * \delta) * \overline{(\beta * \delta)}] = [\alpha * \bar\beta]$.

Step 2. For $\alpha \in \mathcal{P}$ and a path-connected neighbourhood U of $\alpha(1)$ let $B(U,\alpha) = \{(\alpha * \delta)^\# : \delta \text{ a path in } U \text{ from } \alpha(1)\}$; it contains $\alpha^\#$. If $\beta^\# \in B(U,\alpha)$, then $B(U,\beta) = B(U,\alpha)$: with $\beta^\# = (\alpha * \delta)^\#$, elements $(\beta * \gamma)^\# = (\alpha * (\delta * \gamma))^\#$ lie in $B(U,\alpha)$ by (2), and $\alpha^\# = (\beta * \bar\delta)^\#$ gives the reverse inclusion. Hence if $\beta^\# \in B(U_1,\alpha_1) \cap B(U_2,\alpha_2)$, choosing a path-connected $V \subset U_1 \cap U_2$ containing $\beta(1)$, $B(V,\beta) \subset B(U_1,\beta) \cap B(U_2,\beta) = B(U_1,\alpha_1) \cap B(U_2,\alpha_2)$. So these sets form a basis.

Step 3. p is continuous ($B(U,\alpha) \subset p^{-1}(W)$ when $U \subset W$) and open ($p(B(U,\alpha)) = U$).

Step 4. Let U be a path-connected neighbourhood of b_1 with $\pi_1(U,b_1) \to \pi_1(B,b_1)$ trivial. Then $p^{-1}(U)$ is the union of the sets $B(U,\alpha)$ with $\alpha(1) = b_1$: if $p(\beta^\#) \in U$, choose δ in U from b_1 to $\beta(1)$ and $\alpha = \beta * \bar\delta$; then $\beta^\# = (\alpha * \delta)^\#$. Distinct such sets are disjoint by Step 2. And p is injective on $B(U,\alpha)$: if $(\alpha * \delta_1)(1) = (\alpha * \delta_2)(1)$, then $\delta_1 * \bar\delta_2$ is a loop in U at b_1, nulhomotopic in B, so $[\alpha * \delta_1] = [\alpha * \delta_2]$. Being a continuous open bijection, $p|B(U,\alpha)$ is a homeomorphism onto U.

Step 5. Let e_0 be the class of the constant path. For $\alpha \in \mathcal{P}$ and $c \in I$ let $\alpha_c(t) = \alpha(tc)$, and $\tilde\alpha(c) = \alpha_c^\#$. It lifts α from e_0 to $\alpha^\#$. It is continuous at c: a basic neighbourhood of $\tilde\alpha(c)$ is $B(U,\alpha_c)$, U a path-connected neighbourhood of $\alpha(c)$; for d near c, $\alpha([c,d]) \subset U$, and $[\alpha_d] = [\alpha_c * \delta]$ with δ the reparametrized restriction of α to $[c,d]$ (or $[d,c]$, reversed), so $\tilde\alpha(d) \in B(U,\alpha_c)$.

Step 6. E is path connected by Step 5, and locally path connected since it is locally homeomorphic to B; so p is a covering map.

Step 7. For a loop α at b_0, its lifting from e_0 ends at $\alpha^\#$, which equals e_0 iff $[\alpha] \in H$. By Theorem 9.17(c), $p_*\pi_1(E,e_0) = H$. ∎

Corollary 13.16 (82.2)

B has a universal covering space iff it is path connected, locally path connected and semilocally simply connected.

Proof. Necessity: Lemma 13.8 (each point of B is in the image of a point of E).

Sufficiency: Theorem 13.15 with H trivial. ■

Together with Theorem 13.4: for such B, equivalence classes of coverings correspond bijectively to conjugacy classes of subgroups of $\pi_1(B, b_0)$.

13.5 Selected exercises

Motivation. These exercises from Munkres use the general lifting lemma to decide when maps into the circle are nulhomotopic, and give a simple sufficient condition for a group action to be properly discontinuous, so that Theorem 13.13 applies.

Exercise 13.1 (§79, Ex. 1; maps of spheres into the circle)

Show that if $n > 1$, every continuous map $f : S^n \to S^1$ is nulhomotopic.

Solution.

Step 1. S^n is path connected and locally path connected (each point has neighbourhoods homeomorphic to open balls of $\mathbb{R}^n$, by stereographic projection), and simply connected for $n \geq 2$ (Theorem 9.40).

Step 2. Let $p : \mathbb{R} \to S^1$ be the covering of Theorem 9.9, and choose $x_0 \in S^n$ and $e_0 \in p^{-1}(f(x_0))$. Since $f_*(\pi_1(S^n, x_0))$ is trivial, it lies in $p_*(\pi_1(\mathbb{R}, e_0))$, so Lemma 13.1 gives a lifting $\tilde{f} : S^n \to \mathbb{R}$ with $p \circ \tilde{f} = f$.

Step 3. $H(x,t) = p((1-t)\tilde{f}(x))$ is continuous on $S^n \times I$, with $H(x,0) = f(x)$ and $H(x,1) = p(0)$. So f is nulhomotopic. ■

Exercise 13.2 (§79, Ex. 2; maps into the circle)

(a) Show that every continuous map $f : P^2 \to S^1$ is nulhomotopic.
(b) Show that the projection $q : T \to S^1$ of the torus $T = S^1 \times S^1$ onto its first factor is not nulhomotopic.

Solution.

(a) P^2 is a surface (Theorem 9.43), hence path connected and locally path connected. $\pi_1(P^2, x_0)$ has order 2 (Corollary 9.44), so its image under f_* is a finite subgroup of $\pi_1(S^1) \cong \mathbb{Z}$ (Theorem 9.16); the only finite subgroup of $\mathbb{Z}$ is $\{0\}$. Hence f_* is trivial, and Steps 2–3 of Exercise 13.1 apply verbatim with P^2 in place of S^n.

(b)

Step 1. Let $j : S^1 \to T$, $j(z) = (z, b_0)$. Then $q \circ j = \mathrm{id}_{S^1}$.

Step 2. If $H : T \times I \to S^1$ were a homotopy from q to a constant map, then $H \circ (j \times \mathrm{id}_I)$ would be a homotopy from id_{S^1} to a constant, so id_{S^1} would be nulhomotopic.

Step 3. By Lemma 9.20, $(\mathrm{id}_{S^1})_*$ would then be trivial. But it is the identity of $\pi_1(S^1, b_0) \cong \mathbb{Z} \neq 0$, a contradiction. So q is not nulhomotopic. ■

Exercise 13.3 (§81, Ex. 4; free actions of finite groups)

Let X be Hausdorff and G a finite group of homeomorphisms of X acting fixed-point freely (no $g \neq e$ has a fixed point). Show that the action of G is properly discontinuous.

Solution. Let $G = \{g_0 = e, g_1, \dots, g_k\}$ and $x \in X$.

Step 1. The points $g_i(x)$ are distinct: $g_i(x) = g_j(x)$ gives $g_j^{-1} g_i(x) = x$, so $g_j^{-1} g_i = e$ and $i = j$.

Step 2. For $i \neq j$ choose disjoint open sets $W_{ij} \ni g_i(x)$ and $W_{ji} \ni g_j(x)$ (Hausdorff), and let $V_i = \bigcap_{j \neq i} W_{ij}$. The V_i are open, pairwise disjoint, and $g_i(x) \in V_i$.

Step 3. Let $U = \bigcap_{i=0}^{k} g_i^{-1}(V_i)$, an open neighbourhood of x since each g_i is a homeomorphism. Then $U \subset V_0$ and, for $i \geq 1$, $g_i(U) \subset V_i$. As $V_i \cap V_0 = \emptyset$, $g_i(U) \cap U = \emptyset$ for every $g_i \neq e$. ∎